MINISTRY OF EDUCATION, GUYANA

GUYANA NATIONAL GRADE SIX ASSESSMENT

PAPER 02

MATHEMATICS

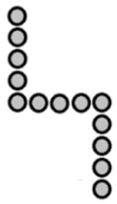
KEY AND MARK SCHEME

2021

MATHEMATICS 2020

KEY AND MARK SCHEME

1. (a) (i) Diagram of **Figure 5**



AT (Attempt made to add 1 button to each side of Figure 4/visually correct)

(ii) **Figure shown:**

- Figure 8 } **K (CAO)**

(iii) **Number of buttons in Figure 10**

$= 3n - 2$

$= 3(10) - 2$ } **R_1 (extending/applying the rule of the pattern/CAO)**

$= 28$

$\therefore$ Figure 10 has 28 buttons

(b) **No. of marbles used:**

Fraction of marbles used = $\frac{2}{3}$

So, fraction left = $1 - \frac{2}{3} = \frac{1}{3}$

Therefore, 14 marbles represent $\frac{1}{3}$ of the total marbles } **R_1** – showing that $\frac{1}{3} = 14$ marbles

Therefore, marbles used to make design

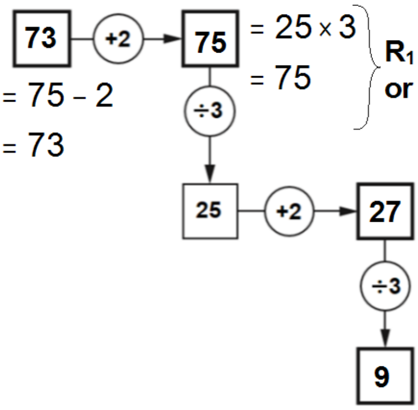
$= 14 \times 2 = 28$ } **AT (multiplying by 2/CAO)**

	K	AT	R	SP OBJ
				CC2.2. CC3.3.
1. (a) (i)		1		
(ii)	1			
(iii)			1	
(b) R_1			1	
(b) AT		1		
TOTAL	**1**	**2**	**2**	**5**

MATHEMATICS 2020

KEY AND MARK SCHEME

Answer	K	AT	R	SP OBJ
2. (a) **Missing values**				**4.3.9.**
$= 25 \times 3 = 75$; $= 75 - 2 = 73$ — **R_1 (having either the 73 or the 75 or both correct)**			1	
$= 25 + 2 = 27$; $= 27 \div 3 = 9$ — **AT_1 (having either the 27 or 9 or both correct)**		1		
(b) (i) **Elements of Sets A and B** Set A = $\{1, 2, 4, 8\}$, Set B = $\{1, 2, 4, 8, 16\}$ — **AT_1 (elements of A or B correct)**		1		4.1.1. 4.1.3.
(ii) **Elements in both sets** $(A \cap B) = \{1, 2, 4, 8\}$ — **K_1 (CAO)**	1			
(iii) **Statement** $A \subset B$ — **R_1 (correct sign used)**			1	
TOTAL	**1**	**2**	**2**	**5**



MATHEMATICS 2020

KEY AND MARK SCHEME

	K	AT	R	SP OBJ
3. (a) (i) **Perimeter of rectangle:** $= 2 \times (\text{Perimeter of the } \Delta)$ $= 2 \times (7 + 7 + 8)\text{cm}$ **R$_1$ (multiplying perimeter of triangle by 2)**			1	**CC8.8** **CC8.9** **CC6.3** **CC6.4**
$= 2 \times 22 \text{ cm}$ $= 44 \text{ cm}$ **K$_1$ (CAO)**	1			
(ii) **Length of rectangle** $\text{Perimeter} = 44 \text{ cm}$ $\therefore 2(l + 9) = 44$ $l + 9 = 22$ $l = 22 - 9$ $l = 13 \text{ cm}$ **AT$_1$ (solving for l)**		1		
(b) (i) **No. of green marbles** $= 50 - \left[\left(\frac{1}{2} \times 50\right) + 15\right]$ $= 50 - (25 + 15)$ **R$_1$ (correct computation)** $= 10 \text{ marbles}$			1	
(ii) **Ratio:** $\text{Red} : \text{blue} = 25 : 15$ $= 5 : 3$ **AT$_1$ (simplified ratio)**		1		
TOTAL	1	**2**	**2**	**5**

MATHEMATICS 2020

KEY AND MARK SCHEME

	K	AT	R	SP OBJ
4. (a) **Responses to statements:**				4.6.3. 4.6.4. 4.6.6. 5.4.3.
(i) A pentagon is a <u>five-sided polygon</u> while a Parallelogram is a <u>four-sided polygon</u>. } $\mathbf{K_1}$ **(CAO)** **N.B. Accept a difference in the number of vertices**	1			
(ii) The <u>square</u> has both parallel and perpendicular lines. } $\mathbf{AT_1}$ **(CAO)**		1		
(iii) The <u>triangle</u> has only acute angles. } $\mathbf{AT_1}$ **(CAO)**		1		
(b) (i) **Angle b** $b = 180^\circ - (98^\circ + 37^\circ)$ $b = 180^\circ - 135^\circ$ $b = 45^\circ$ } $\mathbf{R_1}$ **(subtracting the sum of 98° and 37° from 180°)**			1	
(ii) **Angle a** $a = 360^\circ - (92^\circ + 55^\circ + 58^\circ + 43^\circ)$ $a = 360^\circ - 248^\circ$ $a = 112^\circ$ } **R (adding the 4 angles and subtracting from 360°)**			1	
TOTAL	**1**	**2**	**2**	**5**

MATHEMATICS 2020

KEY AND MARK SCHEME

	K	AT	R	SP OBJ
5. (a) (i) **Selling price of haversack**				CC3.11.
$\text{S.P.} = \text{Cost price} + \text{profit}$ $= \$(3600 + 720)$ $= \$4320$ **AT₁ (adding profit and cost price)**		1		
(ii) **Percentage profit**				
$= \dfrac{\text{profit}}{\text{cost price}} \times 100\%$ $= \dfrac{720}{3600} \times 100\%$ $= 20\%$ **R₁ (multiplying correct fraction by 100%)**			1	
(b) (i) **Number of books**				
$= \dfrac{\$3000}{\$1000} \times 5$ $= 3 \times 5$ $= 15$ **AT₁ (dividing by \$1000 then multiplying by 5)**		1		
OR $1 \text{ book} = \dfrac{\$1000}{5} = \$200$ $\therefore \text{ No. of books} = \dfrac{\$3000}{\$200}$ $= 15$ **AT (dividing by \$200)**				

MATHEMATICS 2020

KEY AND MARK SCHEME

(b) (ii) Extra amount of money needed:

Cost of 100 pencils

$= \frac{100}{4} \times \$100$

$= \$2500$

R_1 (subtracting \$1000 from "his" cost of 100 pencils)

Extra amount needed

$= \$2500 - 1000$

$= \$1500$

K_1 (CAO)

	K	AT	R	SP OBJ
R_1			1	
K_1	1			
TOTAL	**1**	**2**	**2**	**5**

MATHEMATICS 2020

KEY AND MARK SCHEME

6. (a) **Number of passes in science**

- 16

AT_1 (CAO)

(b) **Same number of pupils pass in**

- Mathematics and Reading

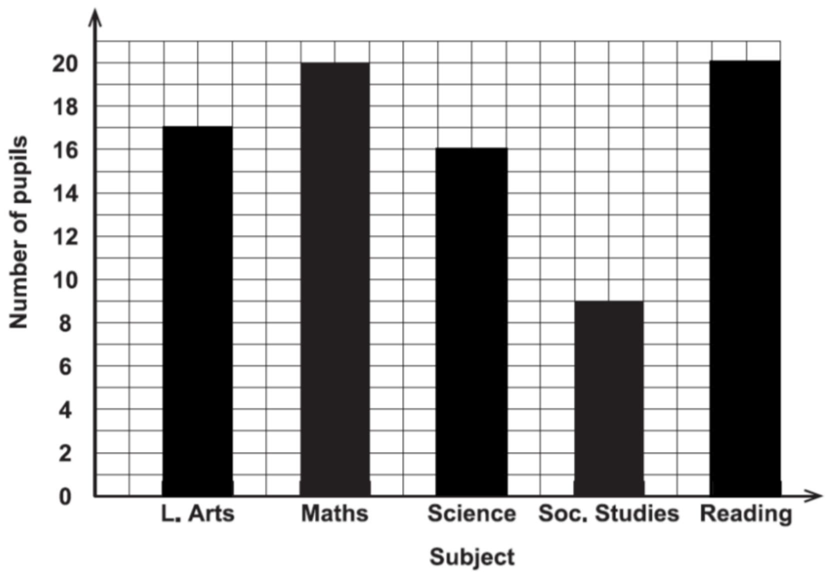
R_1 (CAO)

(c) **Subject with the highest number of pupils who did not pass:**

- Social Studies

R_1 (CAO)

(d) **Completion of graph:**



AT_1 (at most 2 bars correct)

AT_1K_1 (all 3 bars correct)

Item	K	AT	R	SP OBJ
6 (a)		1		CC4.1. CC4.2.
(b)			1	
(c)			1	
(d)	1	1		
TOTAL	**1**	**2**	**2**	**5**